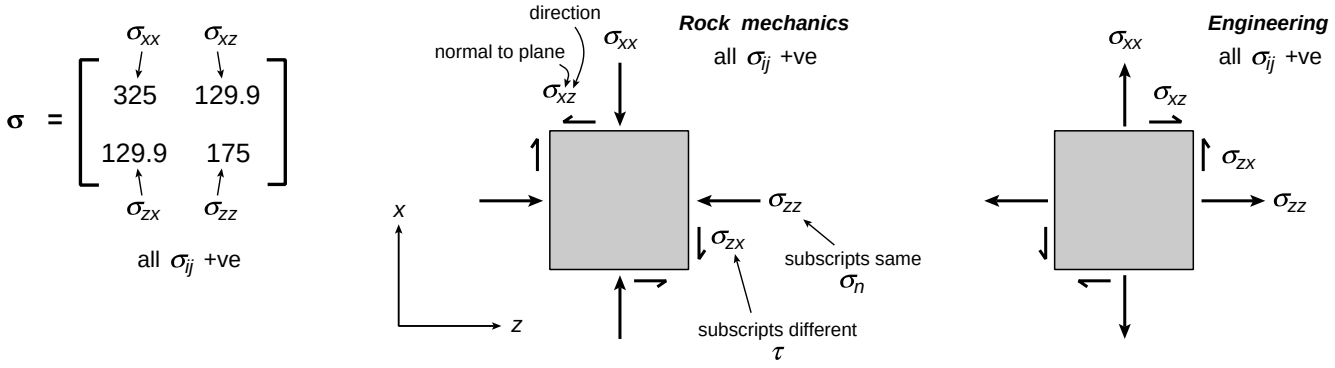
Stress notation and sign conventions

For a 2D state of stress in an x,z coordinate frame:



rock mechanics

compressive normal stresses (σ_n) are +ve

Mohr circle: anticlockwise shear couples (τ) are +ve

engineering

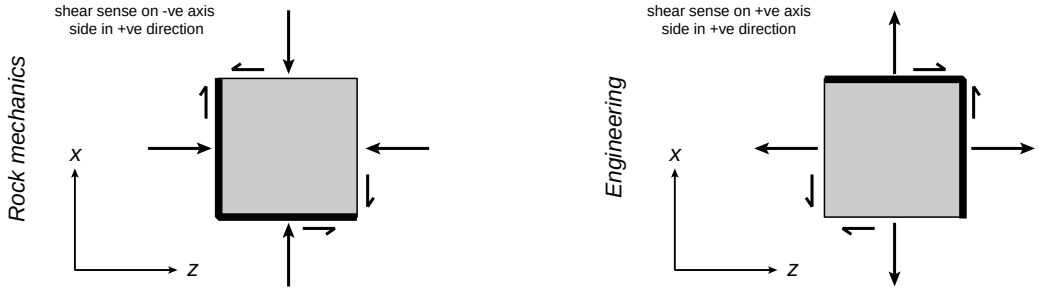
tensile normal stresses (σ_n) are +ve

Mohr circle: clockwise shear couples (τ) are +ve

We will always use the rock mechanics definitions

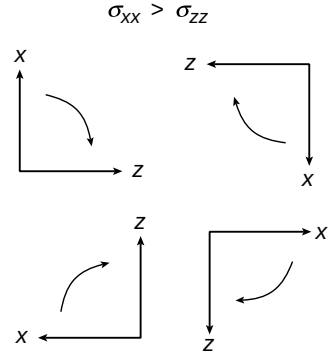
Tensor sign convention (implied when σ_{xz} and σ_{zx} in the stress matrix have the same sign)

The direction of the arrows on the diagram above are set by the tensor sign convention. In rock mechanics this requires you to look at the direction that the stresses point on those sides of your square element (faces of a cube in 3D) that cut the -ve coordinate axes: ***+ve stresses point in a positive coordinate direction on those sides***. In the engineering world, where tensile stresses are +ve and compressive stresses are -ve, the same rule applies except that the stresses on the sides that cut the +ve coordinate axes are considered.



Mohr circle convention (implied when σ_{xz} and σ_{zx} in the stress matrix have different signs)

In the Mohr circle construction one of the shear stresses must be +ve and one must be -ve. In rock mechanics an anticlockwise shear couple is positive. Thus σ_{xz} on the rock mechanics diagrams drawn above is +ve while σ_{zx} is -ve. However, note that a rotation that is anticlockwise when viewed in one direction is clockwise when viewed from the opposite direction. Hence this convention is non-unique − the direction in which you are looking matters. This does not actually matter for 2D stress exercises. It does matter on a 3D Mohr diagram because the three nested Mohr circles on such a diagram must all be viewed in a consistent direction. Consistency is achieved if there is a clockwise sense of rotation from the positive coordinate axis parallel to the greatest normal stress component to the positive coordinate axis parallel to the least normal stress component. If this is not the case, flip the axes. If this convention is followed anticlockwise rotations are +ve on the Mohr diagram.



Consistently oriented frames